



Univerzita Hradec Králové
Filozofická fakulta

E-learningový kurz

Modern quantitative methods
and shape analysis in archaeology



EVROPSKÁ UNIE
Evropské strukturální a investiční fondy
Operační program Výzkum, vývoj a vzdělávání



MINISTERSTVO ŠKOLSTVÍ,
MLÁDEŽE A TĚLOVÝCHOVY

Tento materiál vznikl v rámci realizace projektu
Strategický rozvoj Univerzity Hradec Králové,
reg. č. CZ.02.2.69/0.0/0.0/16_015/0002427.

3D landmark analyses

New 3D ICP-based methods of shape analysis
Practice



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Fully Automated Approach for Aligning and Comparing Shapes

Boyer, D. et al., 2015. A New Fully Automated Approach for Aligning and Comparing Shapes. *The Anatomical Record* 298, 249–276.

A New Fully Automated Approach for Aligning and Comparing Shapes

DOUG M. BOYER,^{1,*} JESUS PUENTE,² JUSTIN T. GLADMAN,^{3,4} CHRIS GLYNN,⁵ SAYAN MUKHERJEE,^{6,7} GABRIEL S. YAPUNCICH,¹ AND INGRID DAUBECHIES⁸

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³NYCEP, New York Consortium in Evolutionary Primatology, New York, New York
⁴Ph.D. Program in Anthropology, Graduate Center, CUNY, New York, New York
⁵Department of Statistical Science, Duke University, Durham, North Carolina
⁶Department of Computer Science, Duke University, Durham, North Carolina
⁷Department of Mathematics, Duke University, Durham, North Carolina

ABSTRACT

Three-dimensional geometric morphometric (3DGM) methods for placing landmarks on digitized bones have become increasingly sophisticated in the last 20 years, including greater degrees of automation. One aspect shared by all 3DGM methods is that the researcher must designate initial landmarks. Thus, if correspondence are required for We present an algorithm allowing for the placement of landmarks on samples of 3D data from individual species, which software and analyzed with the algorithm against several sample datasets represented by 1,024 core automated analysis of these as using a traditional 3DGM approach. Data were analyzed with the returned strong correlations between variance partitioning among shape spaces generated by the cluster analyses of both automated produced broadly similar patterns. These results suggest that the use of shape spaces that are as mesomorphic, thereby presenting potential increase completeness of morphological error, and allow comparisons of bones. We provide an R package, *auto3dgm*, 2015. © 2015



Additional Supporting Information may be found in the online version of this article.
 Grant sponsor: Brooklyn College, CUNY; Duke University and NSF BCS 1317525 to DMB and E. R. Seiffert; NSF 1304045 to DMB and E. M. Seiffert; NYCEP IGERT DGE-0991660, with additional support by a CUNY Grant Center DSRG and Digital Innovations Grant; NYCEP PHROMETRICS GROUP and GSU's work has been supported by the Graduate School of Duke University.

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Fully Automated Approach for Aligning and Comparing Shapes

Why?

- (i) Algorithm can reasonably (!) match corresponding points on different specimens
- (ii) Possible to estimate shape differences
- (iii) allow for the entertainment of different “correspondence hypotheses” based on the morphocline/“path” that is assumed to connect shapes in the dataset

Demonstration on bones

auto3dgm package for R
www.dougboyer.com
<http://www.stat.duke.edu/sayan/>
 CRAN

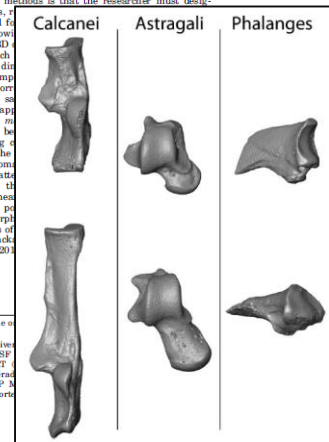
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Concerns with homology

- Bookstein (1994) – morphometrics must be applied after homology consideration
- Boyer et al. (2015) – *auto3dgm* can identify an “operational homology” or “biological correspondence” without consideration *a priori*

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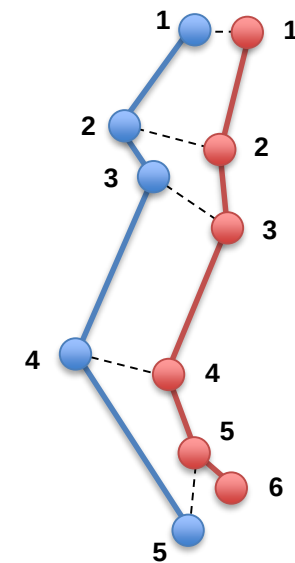


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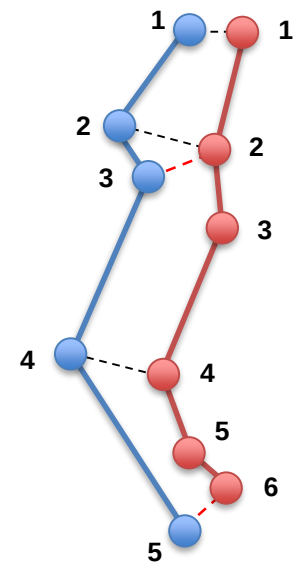
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Iterative Closest Point algorithm (ICP)

- similar to Procrustes - minimize distances between several objects by changing their position, size and rotation
- generalization of Procrustes – can be used to points, but also to lines, splines or surfaces
- difference in the definition of correspondence:
 - Do not use Landmark-to-Landmark...
 - ...but Point-to-Closest-Point
- pseudolandmarks



Procrustes
(Landmark-to-Landmark)



ICP
(Point-to-Closest-Point)

(A) 3D Data preparation

- Low-resolution models for initial alignment
- High-resolution models for final alignment

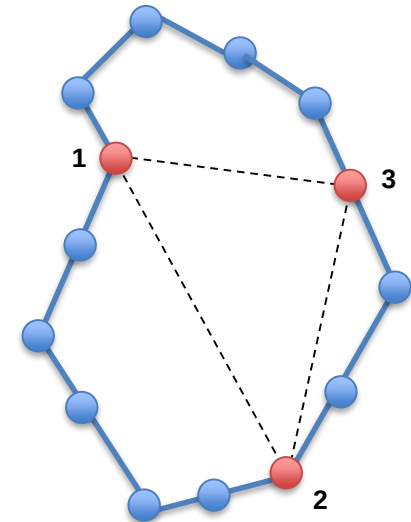
Down-sampling points on meshes

- (1) Randomly pick a point on the surface
- (2) Pick the farthest point
- (3) Pick a third point which is farthest from these two



Grey
Original, high-density mesh

Blue points
"Evenly" spread subsample
for analysis

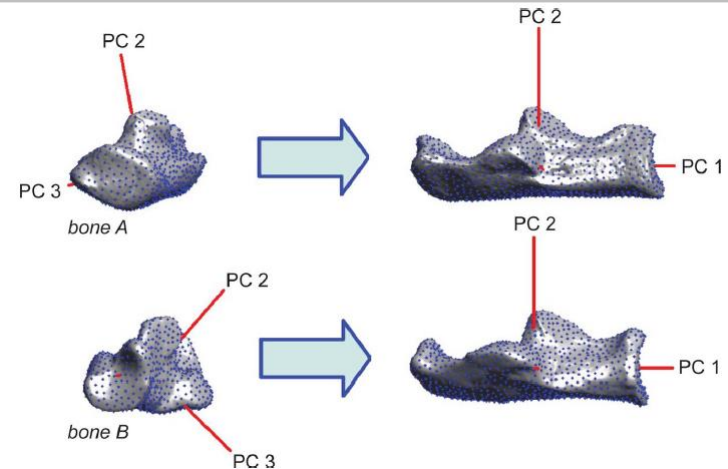


(B) Pre-alignment by PCA

- ICP is known to be sensible to initial guess
- PCA – assumption that principal axes of variation of many 3D objects will tend to be homologous amongst samples
- By taking first three PC there is 8 possibilities of alignment of two objects
- Calculations are made on Low-resolution models

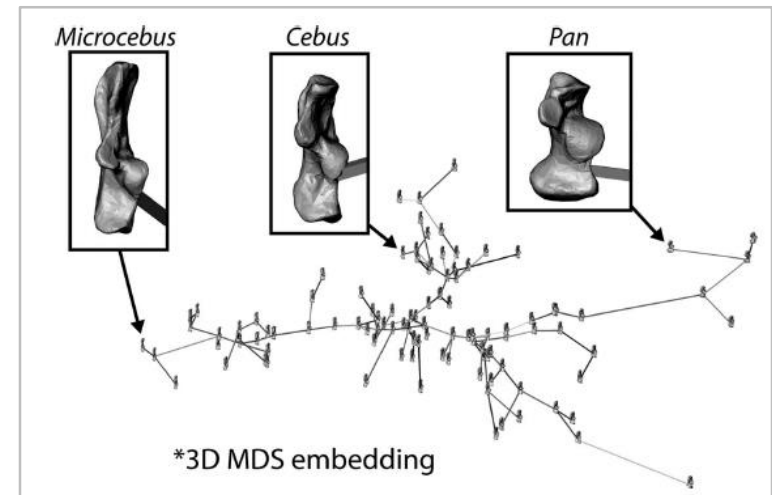
(C) Initial alignment by ICP

- These 8 alignments are taken as initial guesses for ICP
- Calculations are made on Low-resolution models
- Resulting comparisons give matrix of Procrustes distances between specimens

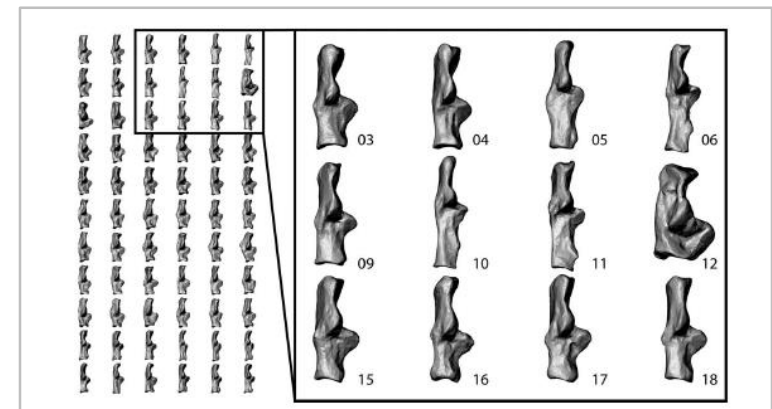


(D) Check initial alignment by MDS and MST

- both methods are calculated on matrix of Procrustes distances
- **Multidimensional Scaling (MDS)** – method analogous to PCA, but more suitable for distance matrices
- **Minimum Spanning Tree (MST)** – connect all specimens by using the shortest edge length possible (nearest neighbors)



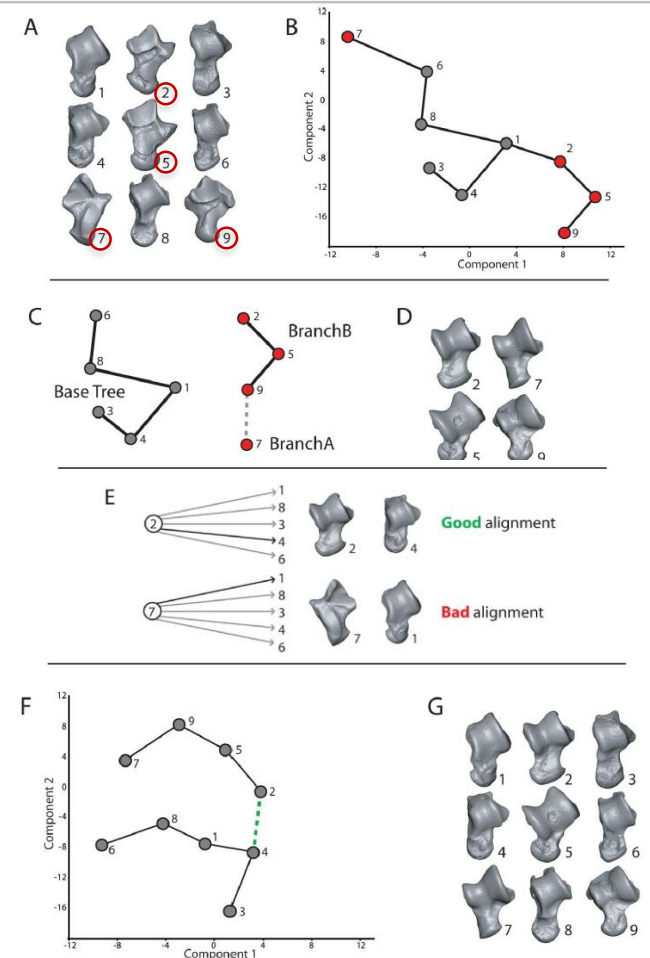
(E) Check initial alignment visually



Fixing alignments (*alignFix* protocol)

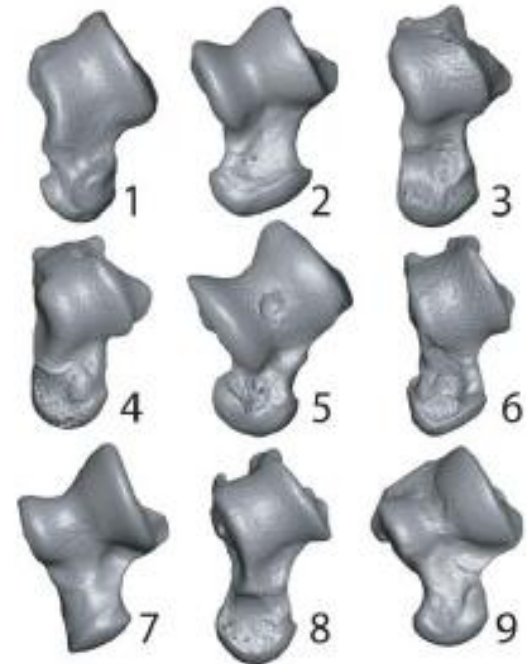
- Identification of misaligned specimens (MS)
- Identification of MS in MST
- Separation of MST into Base tree and Branches
 - Base tree with well aligned specimens
 - Branches with MS (here Branch A and B)
- Test mutual alignment of MS to know if they make one coherent branch (here yes)
- Each MS is aligned with specimens of Base Tree and visually inspected to find joining specimens (here 2 with 4)
- These two specimens are used for connecting coherent branch (obtained in step D) with Base Tree
- Recomputing the alignment (now correct result)

not easy with lot of samples...



(F) Final alignment by ICP

- Calculations are made on High-resolution models
- Only slight variations of position, size, and rotation are used in ICP
- Resulting comparisons give matrix of Procrustes distances between specimens



Parameters

- (1) Number of points for low-resolution first alignment
- (2) Number of points for high-resolution final alignment
- (3) Number of principal alignments
 - 8 by definition
 - but more random principal alignments can be used

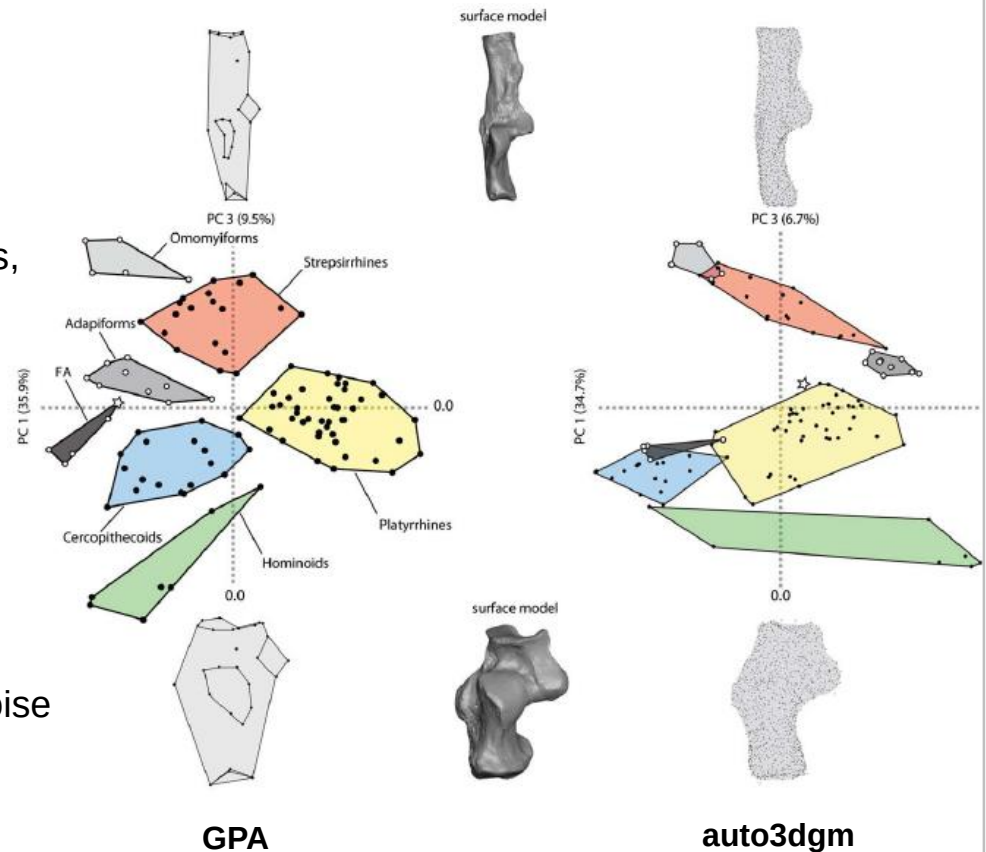
Comparison auto3dgm with GPA

Materials and Methods:

- Dataset: 106 bones, 7 groups
- auto3dgm: Initial = 150 pts, Final = 1024 pts,
Principal alignments = 8
- 3D GPA: 27 landmarks

Results and Discussion

- PCA and correlation between PC showed similarities and coherency
- 3D GPA:
 - + more variance kept by less PC
- auto3dgm
 - more sensitive to errors caused by noise
 - large P, small N problem
 - + no need to assess homology



Visualisation of shape variability of teeth

Question:

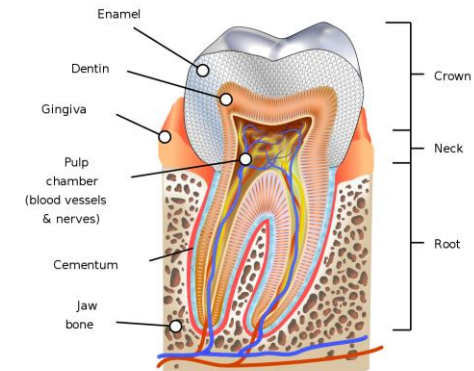
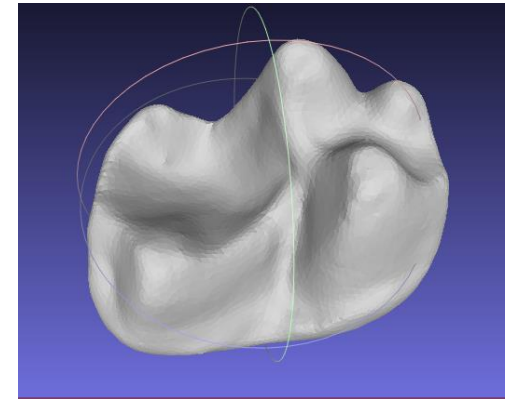
- What are main tendencies related to the shape variability of specimens?

Dataset

- 5 or 15 teeth
- dataset and documentation is available at:
<http://www2.stat.duke.edu/~sayan/auto3dgm/Instructions/index.html>

Procedure

- Perform auto3dgm
- Perform Principal Component Analysis (PCA)
- Perform Principal Coordinate Analysis (PCoA)
- Perform Multidimensional Scaling (MDS)



Software

- Meshlab: www.meshlab.net
- R: <https://www.r-project.org/>
- R Packages: Matrix, clue, IpSolve, linprog, igraph, MASS, geomorph
- PAST: <https://folk.uio.no/ohammer/past/>

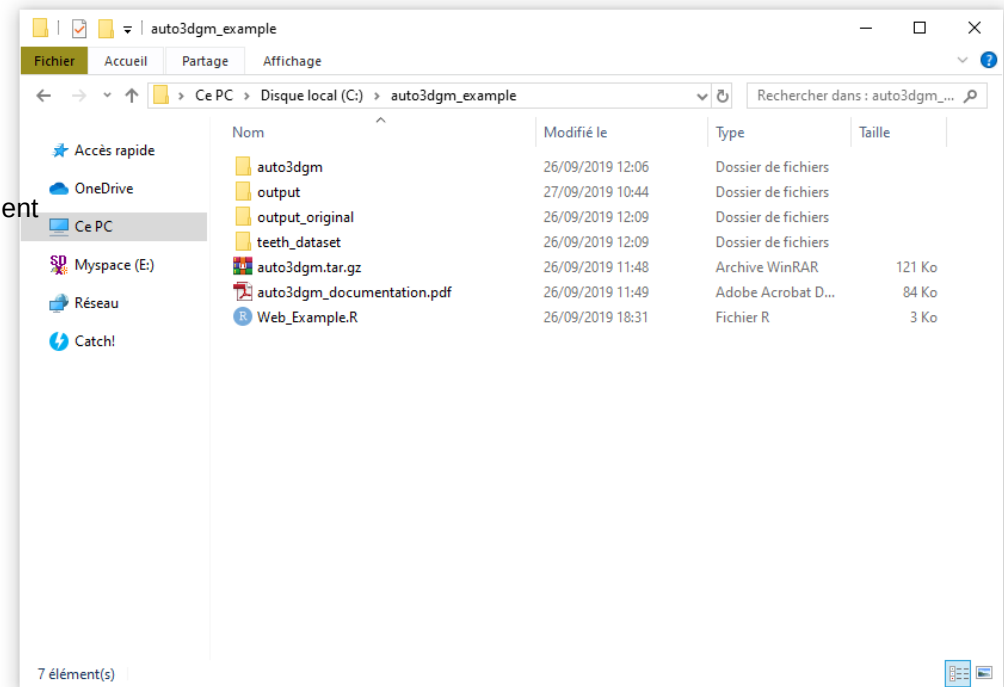


Check folder

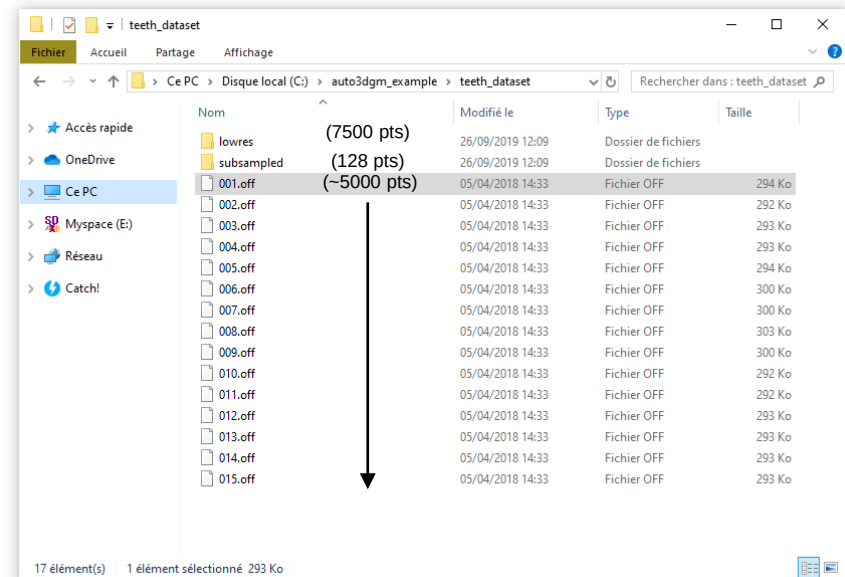
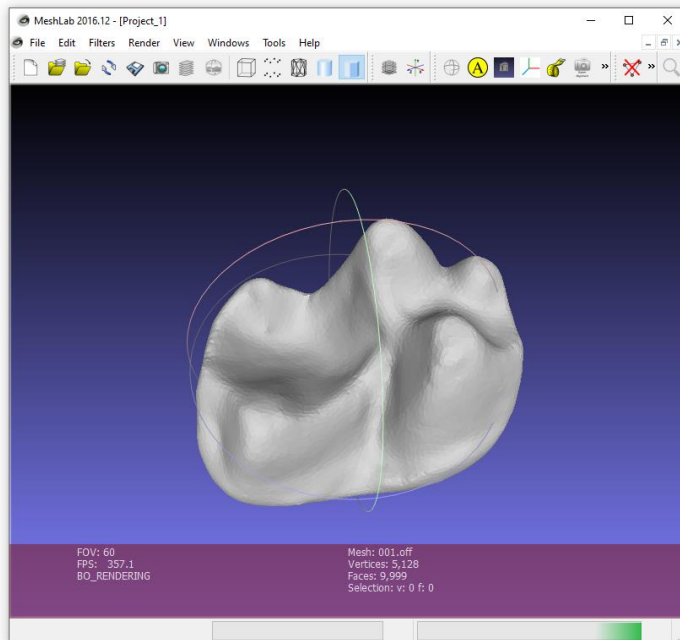
auto3dgm: unpacked package
 auto3dgm.tar.gz: 'auto3dgm' package
 auto3dgm_documentation.pdf: package documentation

teeth_dataset: dataset used in analyses
 output: folder into which we will save results of alignment
 output_original: already calculated results

Web_Example.R: script in R programming language



Check specimens in Meshlab off files in “teeth_dataset” folder



Run procedure in R environment

Read (!), edit (!), copy and paste the code from “Web_Example.R” into R console
See “auto3dgm_documentation.pdf” for more information



```

1 # original documentation is available at: http://www2.stat.duke.edu/~sayan/auto3dgm/instructions/index.html
2 rm(list=ls()) # remove objects from environment
3
4 ## if libraries are not installed, you should run this part
5 # install.packages("matrix")
6 # install.packages("clue")
7 # install.packages("lpsolve")
8 # install.packages("lignprog")
9 # install.packages("igraph")
10 # install.packages("MASS")
11 # install.packages("geomorph")
12
13 ## Load packages
14 library(matrix)
15 library(clue)
16 library(lpsolve)
17 library(lignprog)
18 library(igraph)
19 library(MASS)
20 library(geomorph)
21
22 ## specify path to package folder
23 dir = "C:/auto3dgm_example/auto3dgm/R/"
24 fl1 <- list.files(dir); for (i in 1:length(fl1)) { source(paste(dir,fl1[i],sep="")) }
25
26 ## specify paths to folders
27 data_dir = "C:/auto3dgm_example/teeth_dataset" # folder with files containing 3D models
28 output_dir = "C:/auto3dgm_example/output" # folder with files containing 3D models
29
30 ## specify parameters
31 Levels=c(64,128) # number of points for initial and final alignment
32
33 ## specify individuals you want to analyse
34 ## variante 1 - analyse of 5 individuals
35 ids = c("001", "002", "003", "004", "005") # Ids of 5 individuals
36 Names = c("a01", "a02", "a03", "a04", "a05") # Names of 5 individuals
37 ## variante 2 - analyse of 15 individuals
38 ids = c("001", "002", "003", "004", "005", "006", "007", "008", "009", "010", "011", "012", "013", "014", "015")
39 Names = c("a01", "a02", "a03", "a04", "a05", "a06", "a07", "a08", "a09", "a10", "a11", "a12", "a13", "a14", "a15")
40
41
42 ## run analysis
43 FULL = align_shapes(data_dir, output_dir, Levels, Ids, Names) #FULL is a list of 3 returned elements. The user gets to
44
45 ## check result - each of these three returned items is useful in identifying and fixing misaligned shapes.
46 ds = FULL[[1]] # the whole shape dataset.
47 ga=FULL[[2]] # the global alignment
48 pa=FULL[[3]] # the pairwise alignments.
49
50 ## export results of final alignment to csv file
51 A <- read.morphologika(paste(output_dir,"/morphologika_2.txt",sep="")) # coordinates of all aligned individuals (size
52 shape <- two.d.array(A)
53 write.csv(shape, paste(output_dir,"/coords.csv",sep=""))
54
55 ## you can perform analyses in R or PAST...
56
57

```

Web_Example.R



```

R Console (64-bit)
Fichier Edition Misc Packages Fenêtres Aide

R version 3.6.0 (2019-04-26) -- "Planting of a Tree"
Copyright (C) 2019 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

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Tapez 'contributors()' pour plus d'information et
'citation()' pour la façon de le citer dans les publications.

Tapez 'demo()' pour des démonstrations, 'help()' pour l'aide
en ligne ou 'help.start()' pour obtenir l'aide au format HTML.
Tapez 'q()' pour quitter R.

> |

```

R console

Run procedure in R environment

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24 fil <- list.files(dir); i=1; for (i in 1:length(fil)) { source(paste(dir,fil[i],sep="")) }
25
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31 Levels=c(64,128) # number of points for initial and final alignment
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37 ## variante 2 - analyse of 15 individuals
38 Ids = c('001','002','003','004','005','006','007','008','009','010','011','012','013','014','015')
39 Names = c('a01','a02','a03','a04','a05','a06','a07','a08','a09','a10','a11','a12','a13','a14','a15')
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54
55 ### you can perform analyses in R or PAST...
56
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```

Web_Example.R

```

R Console (64-bit)
Fichier Edition Misc Packages Fenêtres Aide

> library(geomorph)
>
> ### specify path to package folder
> dir="C:/auto3dgm_example/auto3dgm/R/"
> fil <- list.files(dir); i=1; for (i in 1:length(fil)) { source(paste(dir,fil[i],sep="")) }
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> ### specify paths to folders
> Data_dir = "C:/auto3dgm_example/teeth_dataset" # folder with files containi$
> Output_dir = "C:/auto3dgm_example/output" # folder with files containi$
>
> ### specify parameters
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>
> ### run analysis
> FULL = align_shapes(Data_dir, Output_dir, Levels, Ids, Names) #FULL is a list$

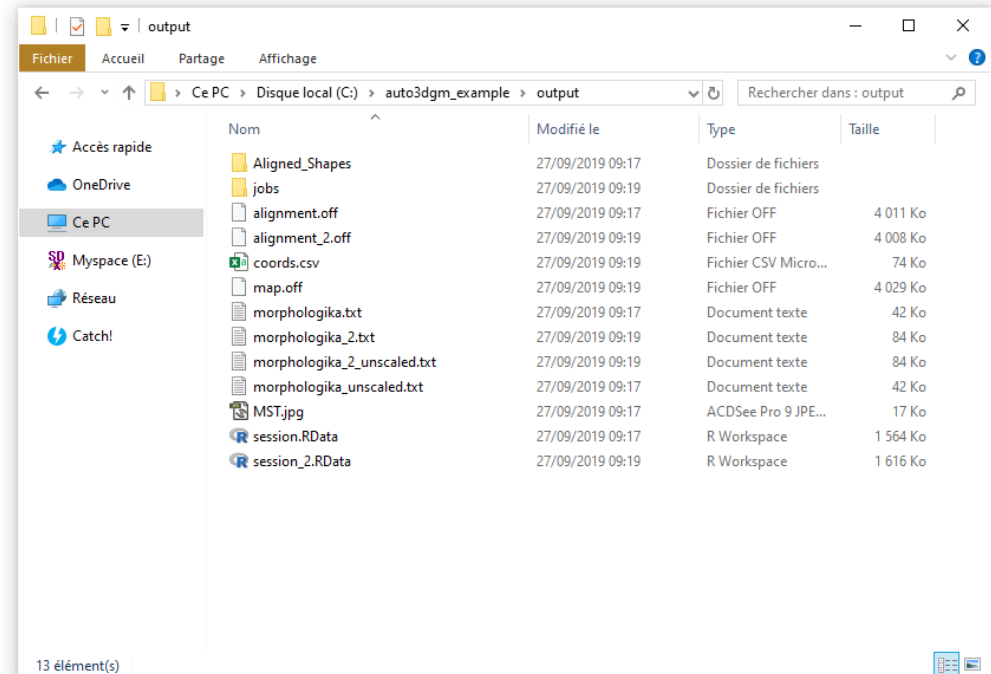
```

CAN TAKE A WHILE...

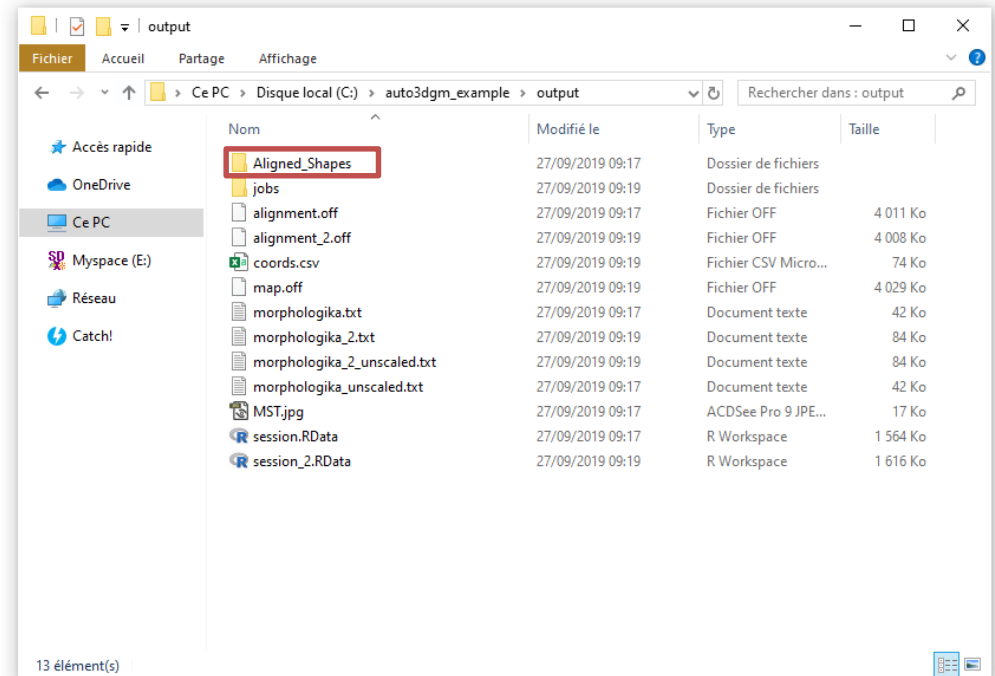
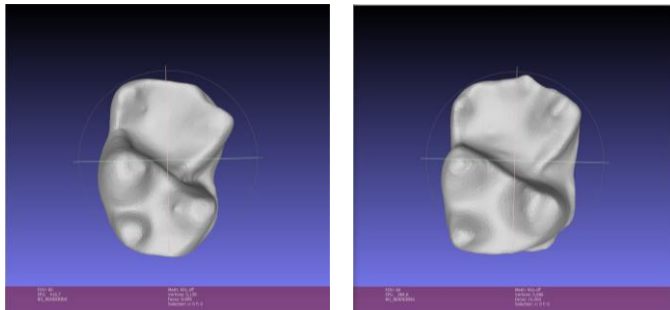
R console

Check results folder

A lot of useful things...

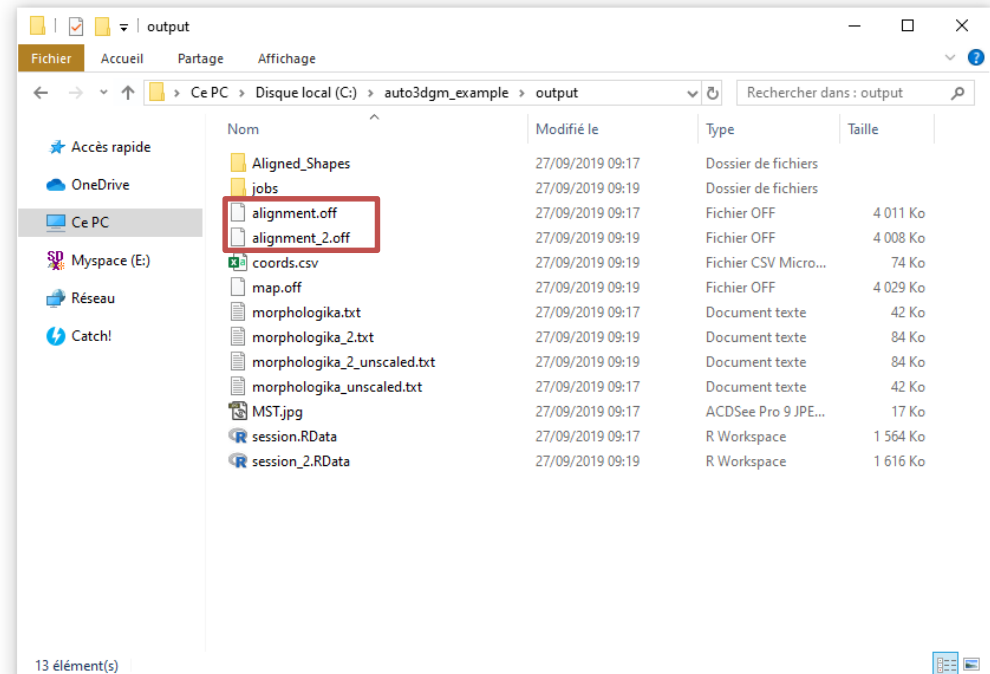
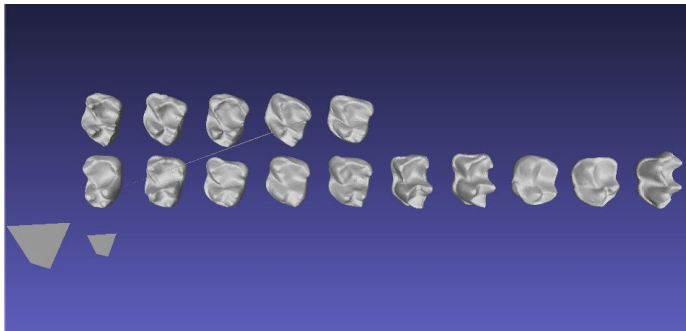
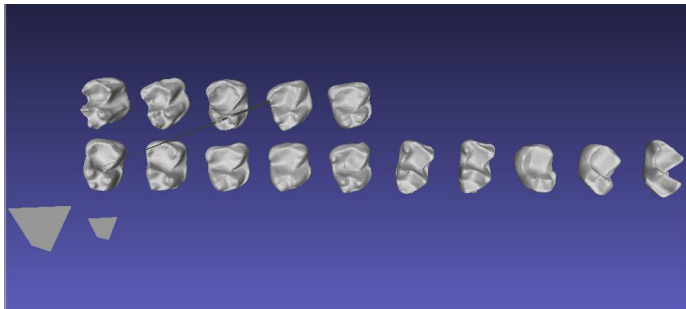


Check results folder Specimens after initial alignment



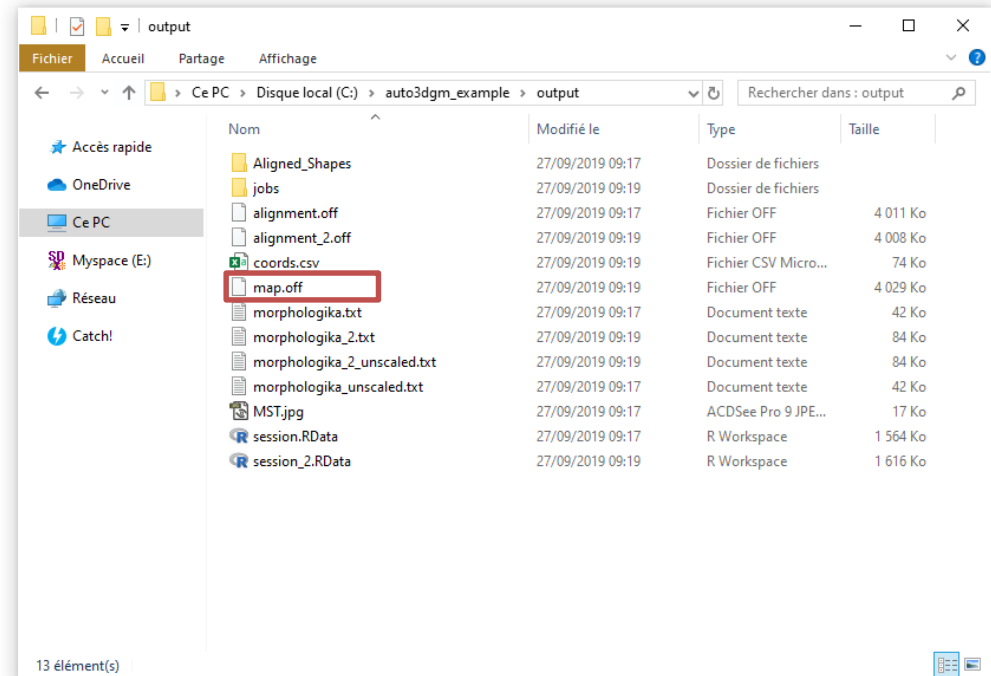
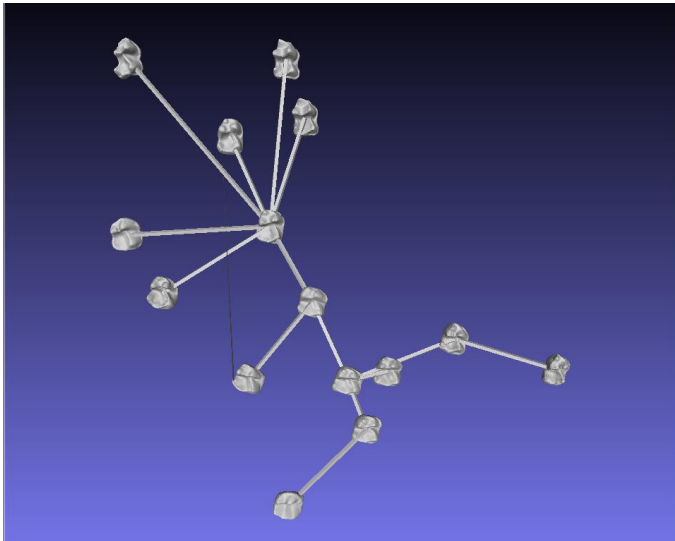
Check results folder

Initial and Final alignments



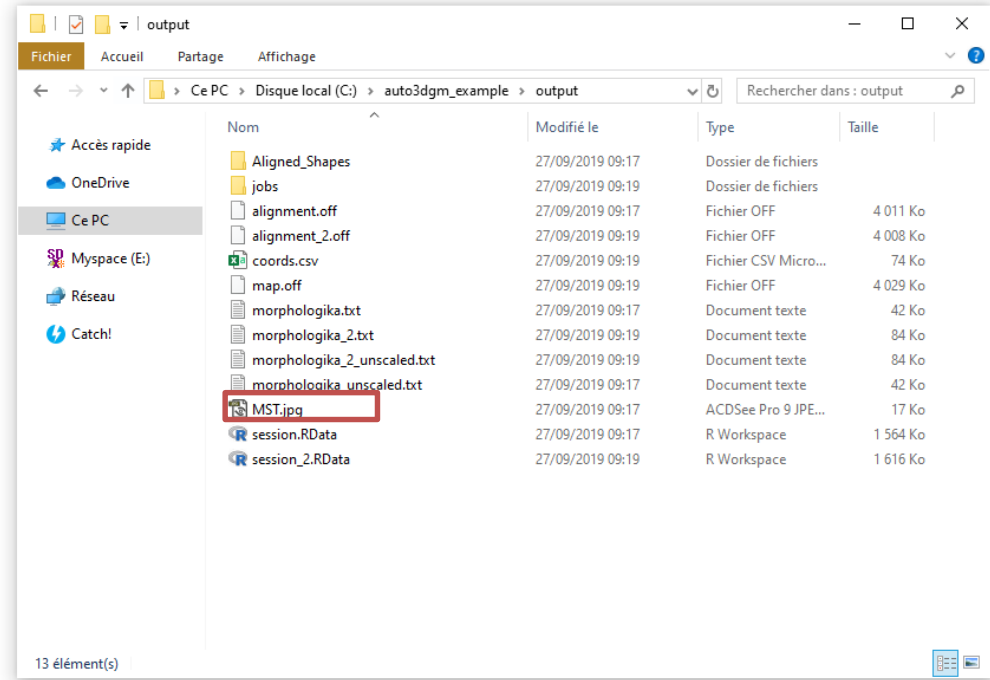
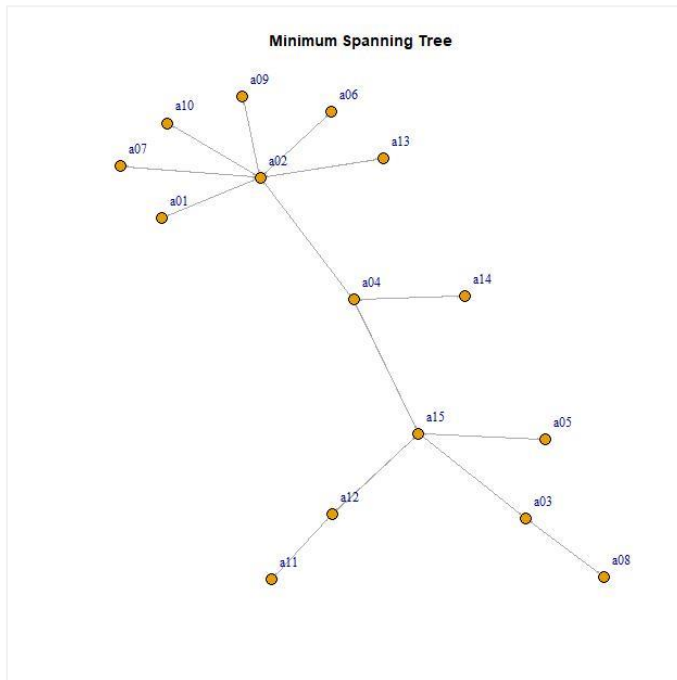
Check results folder

Multidimensional Scaling (MDS) and Minimum Spanning Tree (MST)



Check results folder

Multidimensional Scaling (MDS) and Minimum Spanning Tree (MST)



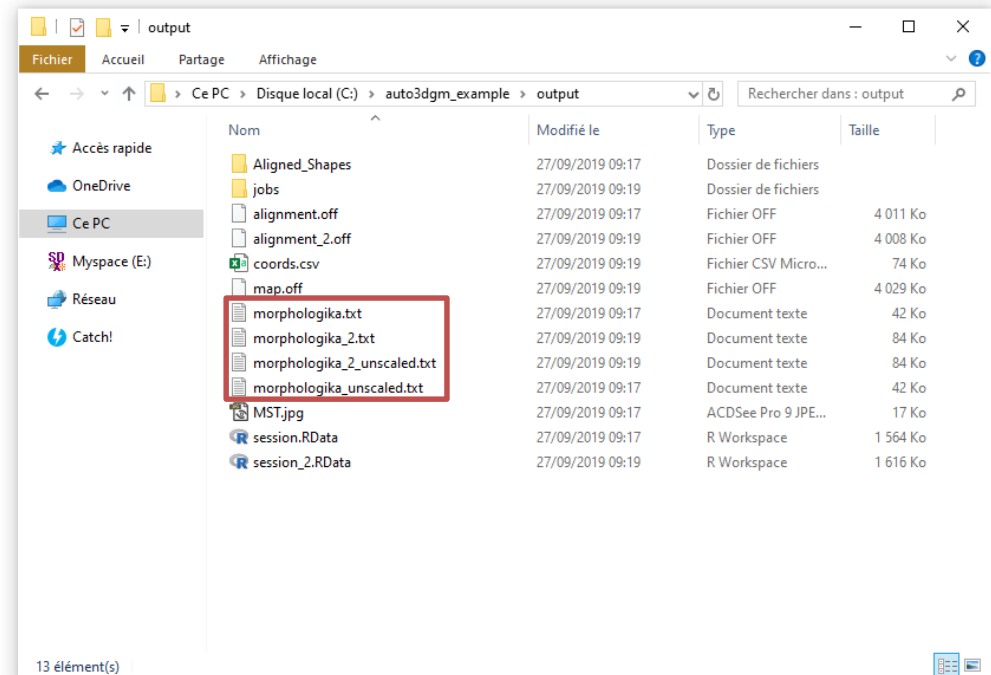
Check results folder

Coordinates of aligned shapes

morphologica – initial alignments

morphologica2 – final alignments

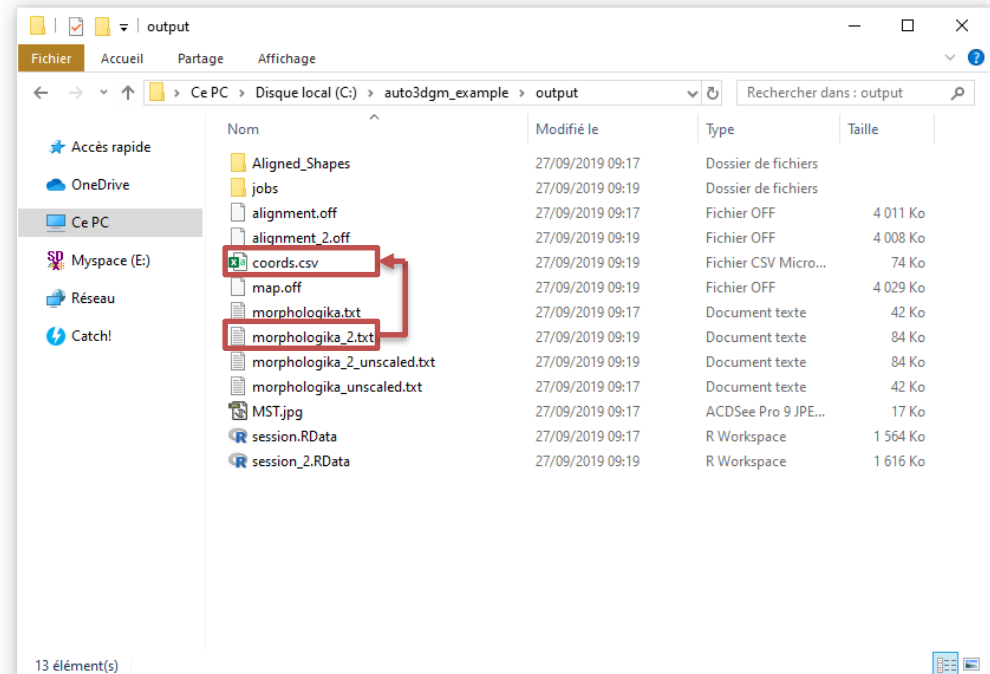
_unscaled – without size adjustment



Check results folder

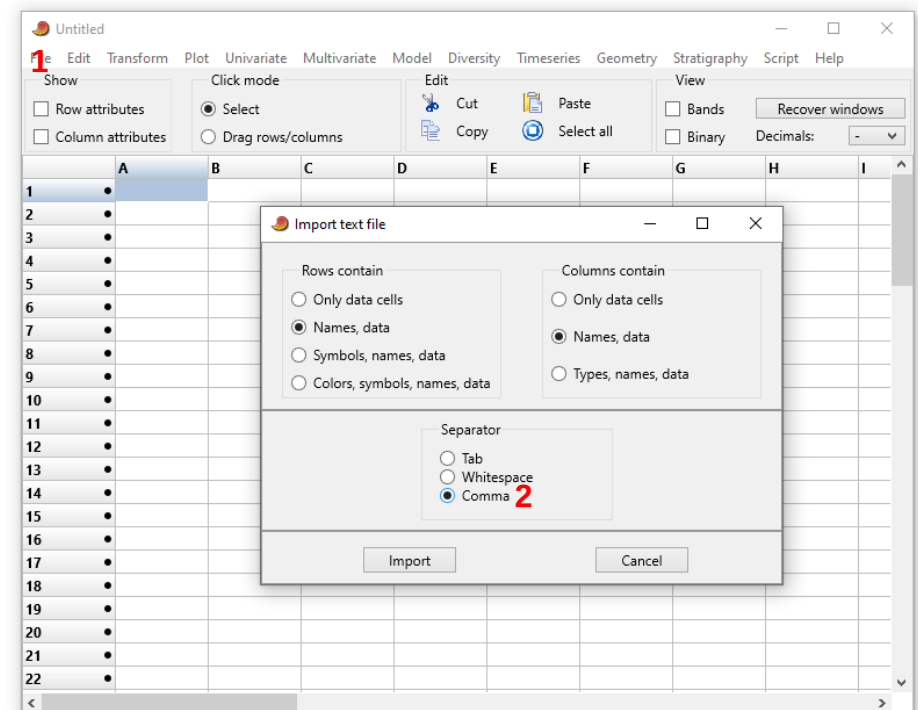
Coordinates of aligned shapes

coords – the same as morphologica2



Data preparation

- 1) Open file:
File >> Open >> cords.csv
- 2) Change Separator to Comma

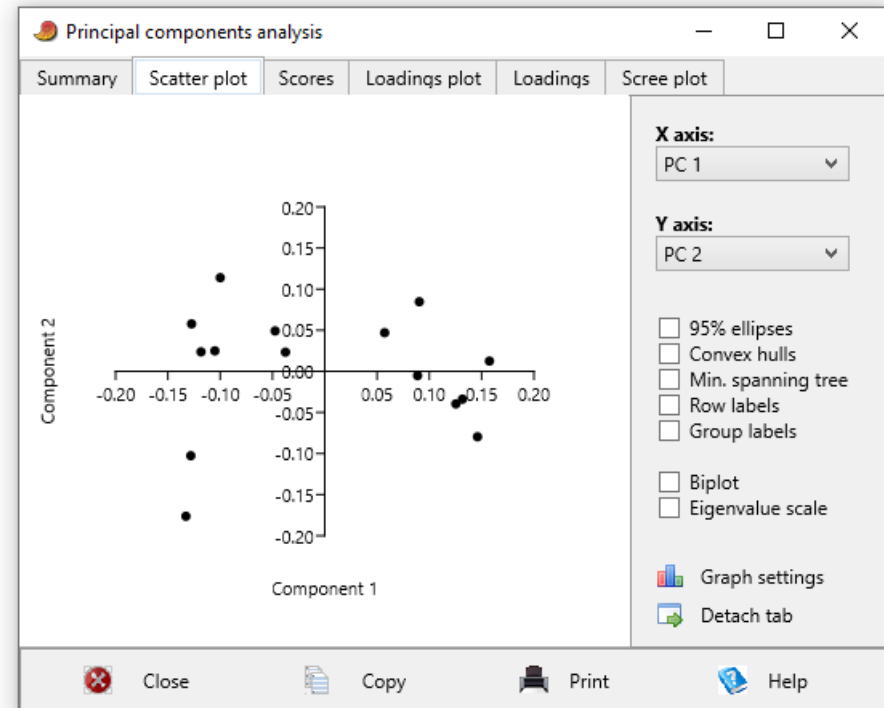


Principal Component Analysis (PCA)

Multivariate >> Ordination >> Principal Components (PCA)

Description and discussion

- how much information we see in graphics?
- how many components are meaningful?
- which changes of shape are represented?

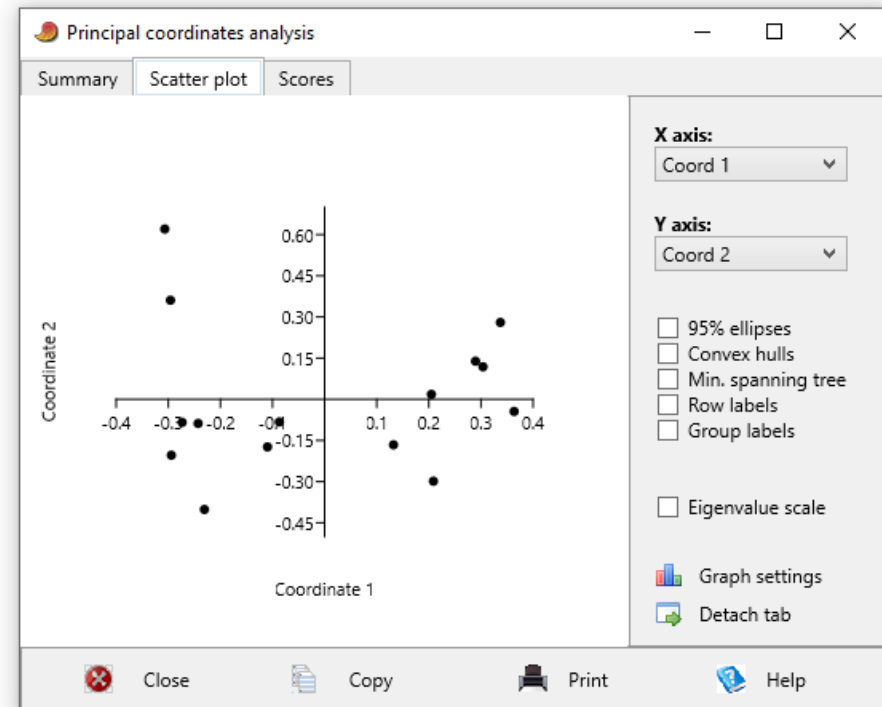


Principal Coordinate Analysis (PCoA)

Multivariate >> Ordination >> Principal Coordinates (PCoA)

Description and discussion

- how much information we see in graphics?
- how many components are meaningful?
- which changes of shape are represented?

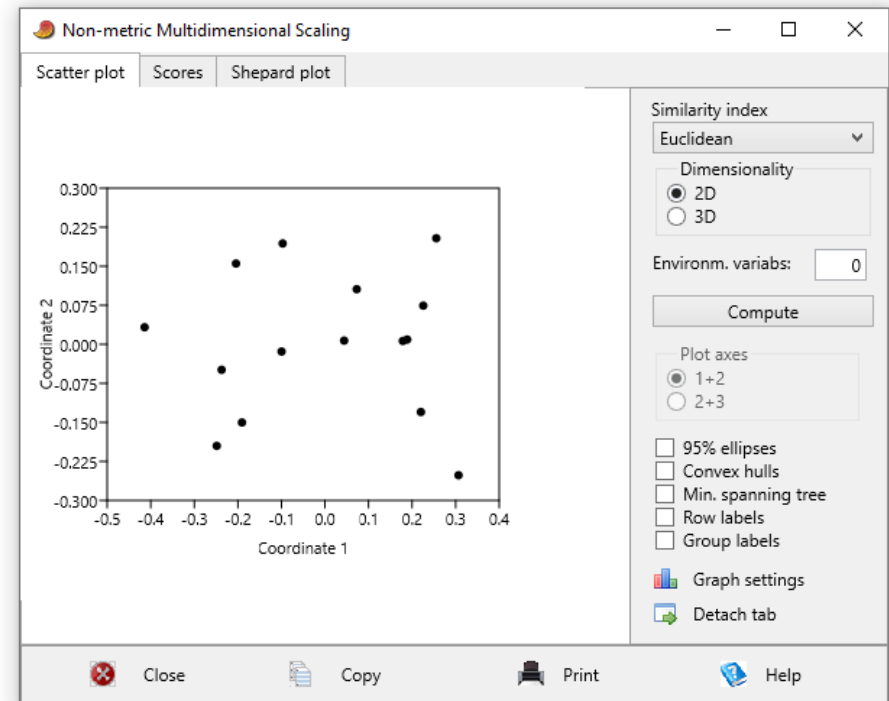


Multidimensional Scaling (MDS)

Multivariate >> Ordination >> Non-metric MDS

Description and discussion

- how much information we see in graphics?
- how many components are meaningful?
- which changes of shape are represented?



References

- Bookstein, F.L., 1994. Can biometrical shape be a homologous character?, in: Hall, B.K. (Ed.), Homology: The Hierarchical Basis of Comparative Biology. Academic Press, San Diego, CA, USA, pp. 197–227.
- Boyer, D.M., Puente, J., Gladman, J.T., Glynn, C., Mukherjee, S., Yapuncich, G.S., Daubechies, I., 2015. A New Fully Automated Approach for Aligning and Comparing Shapes. Anat. Rec. 298, 249–276.
- www.wikipedia.org